FIITJEE

JEE(Advanced)-2018

ALL INDIA TEST SERIES

ANSWERS, HINTS & SOLUTIONS
FULL TEST – I
PAPER-1

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17.	**A**	40.	**A**	63.	**B**
18.	**C**	41.	**D**	64.	**D**
19.	**4**	42.	**4**	65.	**0**
20.	**2**	43.	**5**	66.	**2**
21.	**6**	44.	**5**	67.	**8**
22.	**8**	45.	**6**	68.	**1**
23.	**3**	46.	**4**	69.	**7**

Physics

PART – I

SECTION – A

1. $\overrightarrow{a_B} = \overrightarrow{a_A} + \left(\vec{\alpha} \times \overrightarrow{r_{B,A}}\right) - \omega^2 \overrightarrow{r_{B,A}}$

$= -5j + 4\alpha i + 3\alpha j - (12i - 16j)$

a_B along $y-$axis should be zero.

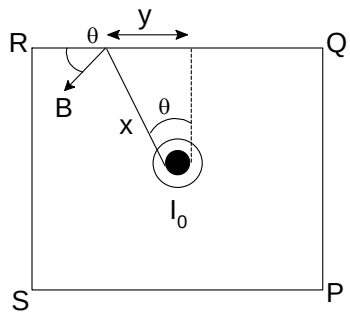
$\Rightarrow 11 + 3\alpha = 0$

$\Rightarrow \alpha = -\frac{4}{3} \text{rad/s}^2$

$a_B = (4\alpha - 12)i = -\frac{80}{3} i \text{ m/s}^2$

2. $y = \frac{\ell}{2}\tan\theta$

$dy = \frac{\ell}{2}\sec^2\theta d\theta$



$8\int_0^{\frac{\pi}{4}} B dy \cos\theta + \oint_{a\to b\to c\to d} \vec{B} \cdot \overrightarrow{d\ell} = \mu_0 I_0$

$\int_0^{\frac{\pi}{4}} \frac{8\mu_0 I_0}{4\pi x} \cdot \frac{2\frac{\ell}{2}}{\sqrt{x^2 + \frac{\ell^2}{4}}} \cdot \frac{\ell}{2}\sec^2\theta\cos\theta + \oint_{a\to b\to c\to d} \vec{B}.\overrightarrow{d\ell} = \mu_0 I_0$

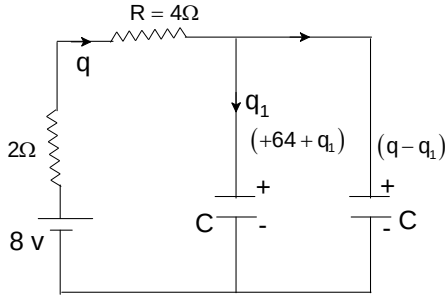
$\frac{2}{3}\mu_0 I_0 + \oint_{a\to b\to c\to d} \vec{B}.d\ell = \mu_0 I_0$

$\oint_{a\to b\to c\to d} \vec{B}.\overrightarrow{d\ell} = \mu_0 I_0 - \frac{2}{3}\mu_0 I_0 = \frac{\mu_0 I_0}{3}$

3. Let $R + 2 = R_1$

$q(t) = \frac{3CE}{2}\left(1 - e^{-\frac{t}{2R_1C}}\right)$

$I = \frac{dq}{dt} = \frac{3E}{4R_1} e^{-\frac{t}{2R_1C}}$



Heat loss in $R_1 = \int I^2 R_1 dt$

$= 576\mu J$

so, heat loss in $R = 4\Omega$ is $= 576 \times \frac{2}{3} = 384\mu J$

4. Maximum power transfer will takes place when v_P is maximum.

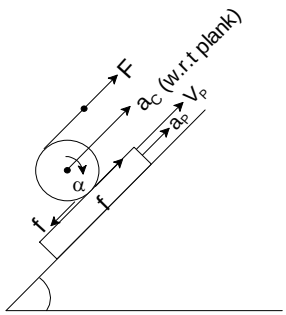
5. $a_C = R\alpha$ (for no slipping)

$V_C = R\omega$ $(V_C$ is the speed of cylinder w.r.t plank$)$

$a_C + 3a_P = 0$

$V_C + 3V_P = 0$

$S_C + 3S_P = 0$



6. Let T_1 and T_2 be the initial and final temperature, then

$P_0V = nRT_1$ .....(i)

$\left(P_0 + \frac{W}{A}\right)(V - Ah) = nRT_2$ ....(ii)

Where h is the displacement of piston.

Also, work done on gas $= \Delta U$

$Wh = nC_V\Delta T = \frac{3}{2}nR(T_2 - T_1)$ .....(iii)

From euqation (i), (ii) and (iii)

$\frac{2}{3}Wh = \frac{WV}{A} - Wh - P_0Ab$

or, $Ah = \frac{WV}{P_0A + \frac{5}{3}W}$

$T_2 = \left(P_0 + \frac{W}{A}\right)\left(V - \frac{Wv}{P_0A + \frac{5}{3}W}\right)$

$T_2 = \frac{5}{4}T_1 = \frac{5}{4} \times 300 = 5 \times 75K.$

7. $f = \frac{v}{2(\ell + 2 \times 0.6r)} = \frac{v}{2(\ell + 1.2r)}$

$\therefore \frac{\Delta f}{f} = \frac{\Delta v}{v} - \frac{\Delta(\ell + 1.2r)}{(\ell + 1.2r)}$

Here $\frac{\Delta v}{v} = 0$

$\frac{\Delta f}{f} \times 100 = -\frac{(\Delta\ell + 1.2\Delta r)}{(\ell + 1.2r)} \times 100$

$= 0.16$

8. $\Delta P = \frac{Gm^2}{8\pi R^4}$

$$B = -\frac{\Delta P}{\Delta V / V}$$

$$B.\frac{3dR}{R} = \frac{Gm^2}{8\pi R^4}$$

$$dR = \frac{Gm^2}{24B\pi R^3}$$

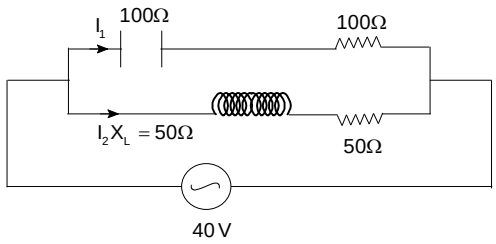
9. $I_1 = \frac{20}{z_1} \times 2$

$= \frac{2}{5\sqrt{2}}$ A at 45^0 leading

$I_2 = \frac{20}{z_2} = \frac{\sqrt{2}}{5}$ A at 45^0 lagging

$I_{1_{R.M.S}} = \frac{1}{5}$ A

$I_{2_{R.M.S}} = \frac{2}{5}$ A



10. For central height fringe

$d\sin\theta = 2mm$

$4mm\ \sin\theta = 2mm$

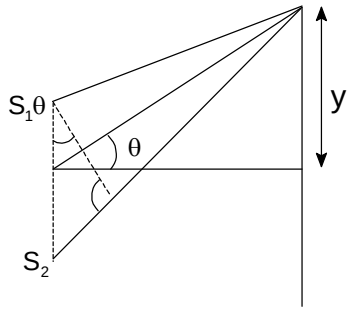
$\theta = 30^0$

$\tan 30^0 = \frac{y}{2}$

$y = \frac{1}{\sqrt{3}}$m.

For bright fringe $(d\sin\theta \pm 2) = n\lambda$

where n is 0,1,2,3.....



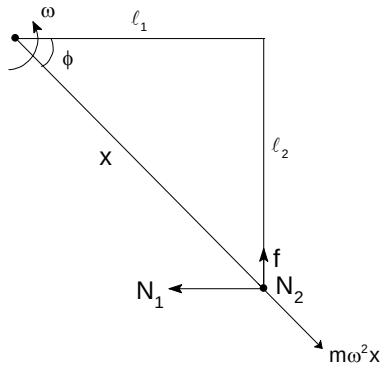
11. $N_1^2 + N_2^2 = N^2$

$N_2 = mg$

$m\omega^2 \times \sin\phi = f$

$f = m\omega^2 \ell_2$

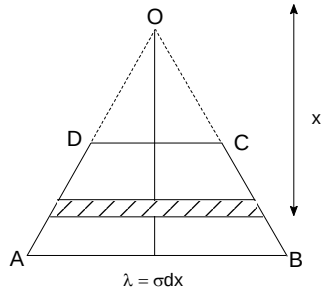
$N_1 = m\omega^2 \ell_1$



12. $dE = 2\left(\frac{1}{4\pi\varepsilon_0}\right)\frac{\sigma dx}{x}\left(\frac{1}{2}\right)$

$dE = \frac{\sigma}{4\pi\varepsilon_0}\frac{dx}{x}$

$E = \frac{7\sigma}{44\varepsilon_0}\ln\sqrt{2}$



13. & 14.

Writing energy equation

$mgv = mv\frac{dv}{dt} + I\frac{\omega d\omega}{dt} + i^2R \quad .....(i)$

$v = a\omega$

$\frac{dv}{dt} = a\frac{d\omega}{dt}$

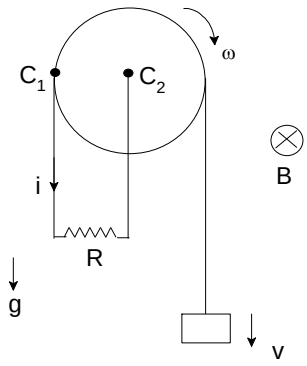
$i = \frac{B\omega a^2}{2R} = \frac{Bva}{2R} \quad .....(ii)$

From equation (i) and (ii)

$\frac{3m}{2}\frac{dv}{dt} = mg - \frac{B^2a^2}{6mR}v$

$v = \frac{4mgR}{B^2a^2}\left(1 - e^{-\left(\frac{B^2a^2}{6mR}\right)t}\right)$

$\frac{dv}{dt} = \frac{2g}{3}e^{-\frac{B^2a^2}{6mR}t}$



15. & 16.

$V_{emit} > V_{collector}$

$V = -iR$

$V_{stop} = \frac{hc}{e\lambda} - \frac{\phi}{e} = 2v$

$i = 2 + V$

$I = 2 - iR$

$i(1+R) = 2$

$i = \frac{2}{1+R}$

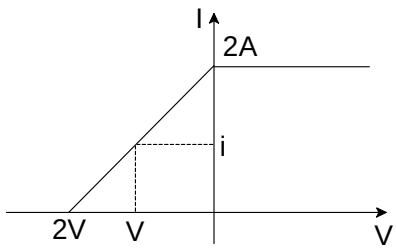
$P = i^2R = \frac{4R}{(1+R)^2}$

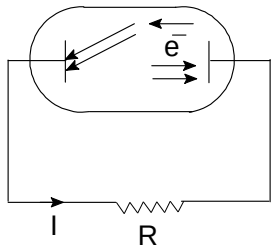
$\frac{dP}{dR} = 0$

$4(1+R)^2 - 4R \times 2(1+R) = 0$

$4 + 4R = 8R$

$R = 1\Omega$





17. & 18.

$V = \sqrt{2gh} = 1\,m/s$

$-\frac{8}{100} = \frac{3}{5}t - \frac{1}{2}\times 10Xt^2$

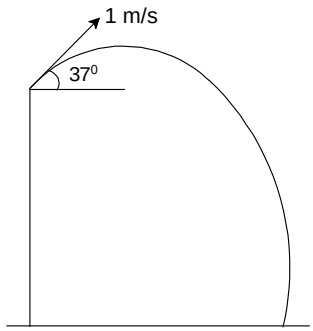
$t = \frac{1}{5}sec$

$\therefore x = \frac{4}{5}\times\frac{1}{5}\times 100 = 16\ cm$

horizontal range from point $B = 16 - 6 = 10\,cm$

Force required to keep the vessel in equlibrium $= \rho AV^2\cos 37^0$

$= 0.12\ N$



SECTION – C

19. $FR - \frac{\mu_0 q^2 R^2}{4\pi\ell}\alpha = mR^2\alpha$

$F = R\alpha\left[m + \frac{\mu_0 q^2}{4\pi\ell}\right]$

$\frac{F}{2m} = R\alpha = \frac{F}{m + \frac{\mu_0 q^2}{4\ell\pi}}$

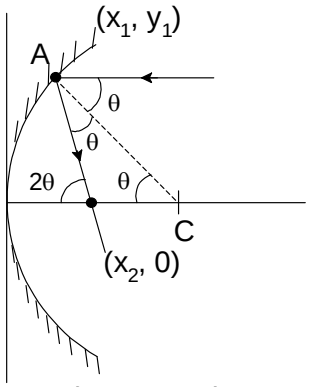
$q = \sqrt{\frac{m\ell 4\pi}{\mu_0}} = 5\times 10^4$

20. After refraction through the quarter cylinder the emergent ray falls on the parabolic reflector parallel to x – axis.

$\tan(\pi - \theta) = -\frac{1}{\left.\frac{dy}{dx}\right|_{(x_1,y_1)}} = y_1$

$\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta} = \frac{y_1}{x_2 - x_1}$

$\Rightarrow x_2 = \frac{1}{2}m = 50\ cm.$



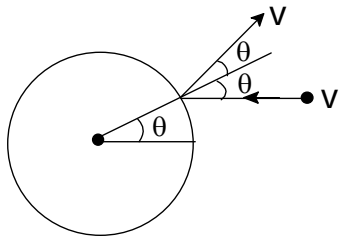
Ac is a normal at A

21. $\frac{1}{2}.\frac{m}{2}v_0^2 = \frac{1}{2}K\left(\frac{3F}{K}\right)^2 + \frac{3F}{2}.\frac{3F}{K}$

$v_0^2 = \frac{4\times 9F^2}{Km}$

$v_0 = 6\ m/s$

22. In the frame of the heavy cylinder the particle comes in with speed V and then bounces off.



$$f = \int_{-\frac{\pi}{2}}^{+\frac{\pi}{2}} 2mv\cos^2\theta\left(Rd\theta\cos\theta\right)\ell vn$$

$$\frac{f}{\ell} = \frac{8}{3}mv^2nR.$$

23. $f = f_0\frac{z^2}{n^3} \Rightarrow \frac{z^2}{n^3} = 2 \quad .....(i)$

$E = E_0\frac{z^2}{n^2} \Rightarrow \frac{z^2}{n^2} = 4 \quad (ii)$

Solving (i) and (ii) n = 2 z = 4

$L = mvr = \frac{nh}{2\pi}$

$\Delta L = \tau\Delta t = \Delta n\frac{h}{2\pi}$

$\tau = \frac{\Delta n}{\Delta t}.\frac{h}{2\pi}$

$\tau \times 10^{27} = 15$

Chemistry

PART – II

SECTION – A

24. **(C)**
Fact based

25. **(D)**
Zone refining is based on difference in solubility of impurity in its molten as well as solid state.

26. **(C)**

No. of moles of AgCl = $\frac{1.435\ gm}{M_{AgCl}} = 0.01\ moles$

No. of moles of $CoCl_3.4NH_3$ = 0.01 moles

$\therefore$ 1 ionizable Cl per mole of $CoCl_3.4NH_3$

27. **(B)**

$AgI(s) \rightleftharpoons Ag^+(aq) + I^-(aq)$

$K_{sp} = \left[Ag^+(aq)\right]\left[I^-(aq)\right]$

$Ag^+(aq) + 2CN^-(aq) \rightleftharpoons \left[Ag(CN)_2\right]^-_{(aq)}$

$K_f = \frac{\left[Ag(CN)_2\right]^-}{\left[Ag^+\right]\left[CN^-\right]}$

Now, let a moles of AgI dissolved in CN^- solution

$$\underset{1\,mole}{AgI(s)} + \underset{(1-2x)\,mole}{2CN^-_{(aq)}} \rightleftharpoons \underset{x\,mole}{\left[Ag(CN)_2\right]^-_{(aq)}} + \underset{x\,mole}{I^-_{(aq)}}$$

As $\left[CN^-\right] = M.V = 1\times 1 = 1\,mole$

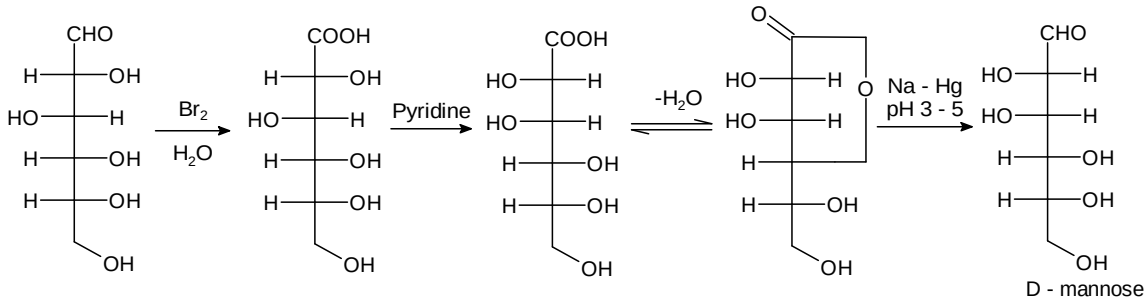
$\therefore K_c = \frac{x.x}{(1-2x)^2}$

also $K_c = K_f.K_{sp}$

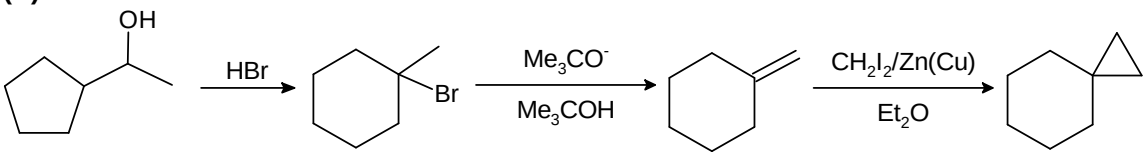
$\therefore \frac{x^2}{(1-2x)^2} = 1.19\times 10^{-17}M^2 \times 7.11\times 10^{19}M^{-2}$

$\therefore x = 0.5$

28. **(A)**



29. **(D)**



30. **(A)**

$-\frac{d[A]}{dt} = k.[A]$; *for first order reaction*

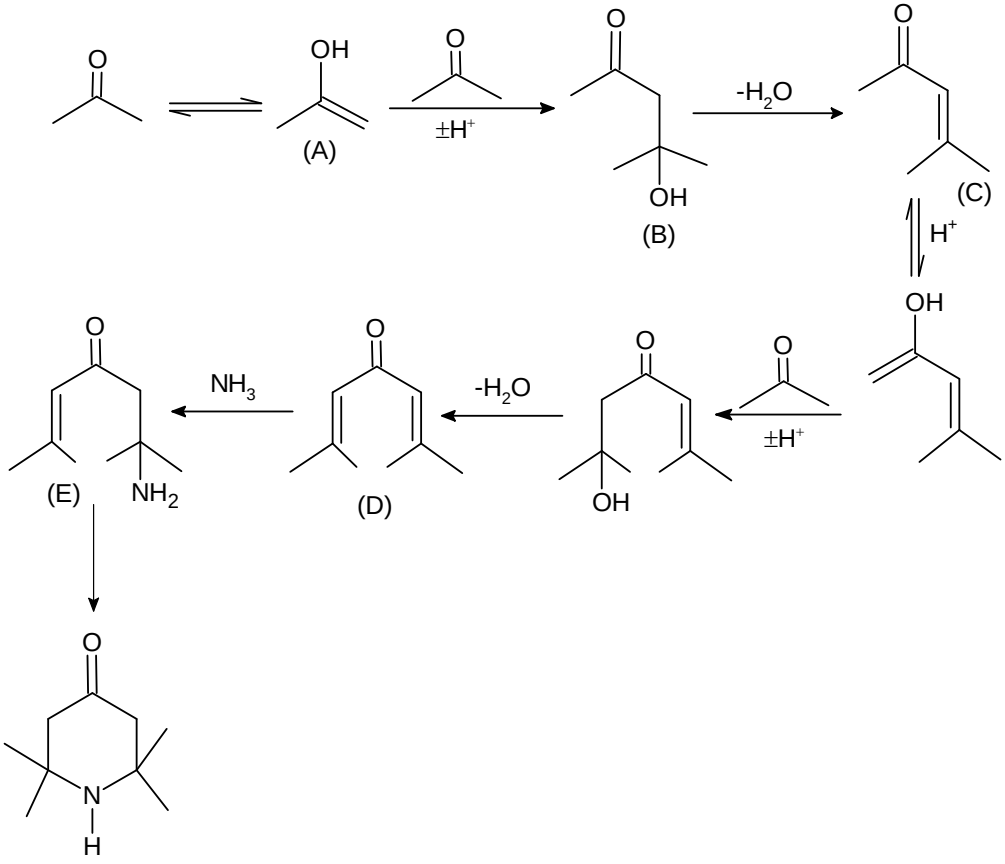
$[A] = [A]_0 - kt$; *for zero order reaction*

$\frac{1}{[A]_t} = \frac{1}{[A]_0} + kt$ *for* sec*ond order reaction*

31. **(A)**

$2KMnO_4 + KBr + H_2O \longrightarrow KBrO_3 + 2KOH + 2MnO_2$

32. **(A, B)**



33. **(A, B, C)**

$\Delta U = nC_V \Delta T \ (\because \Delta T = 0; \Delta U = 0)$

$\Delta H = nC_P \Delta T \ (\because \Delta T = 0; \Delta H = 0)$

1st *law* $\Delta U = q + w = 0$

$\therefore q = -w$

$w = -\int_{V_1}^{V_2} P_{ext}\, dv$

$P_{ext} = P_2; w = -P_2(V_2 - V_1)$

$w = -P_2\left[\frac{RT}{P_2} - \frac{RT}{P_1}\right]$

$= -RT\left[1 - \frac{P_2}{P_1}\right]$

$\therefore q = RT\left[1 - \frac{P_2}{P_1}\right]$

34. **(B, D)**

$P_C = \frac{1}{27}\frac{a}{b^2}$

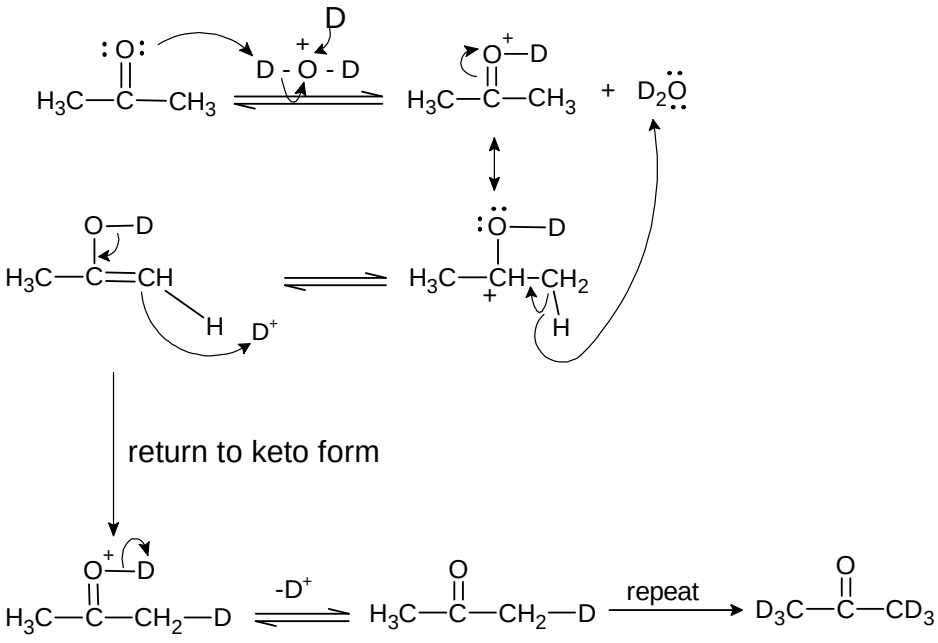
$V_C = 3V_b$

$T_C = \frac{8}{27}\frac{a}{Rb}$

35. **(B, C)**

As for salt of strong acid and weak base degree of hydrolysis, $h = \sqrt{\frac{K_w}{K_b.C}}$

36. **(A)**



37. **(B)**

As deprotonation step in case of iodination is rate determining.

38. **(C)**

$$E_{cell} = E^{o}_{cell} - \frac{RT}{nF}\ln Q$$

$$E_{cell} = 0 - \frac{RT}{nF}\ln\left[\frac{C_2}{C_1}\right]$$

$$= -\frac{RT}{nF}\ln\left[\frac{\frac{1}{50}}{\frac{2}{100}}\right]$$

$$= 0V$$

39. **(A)**

oxidation half – cell reaction

$Ag(s) \rightarrow Ag^{+}(aq) + e^{-}$

Reduction half – cell reaction

$AgCl(s) + e^{-} \longrightarrow Ag(s) + Cl^{-}(aq)$

Net cell Reaction : –

$AgCl(s) \longrightarrow Ag^{+}(aq) + Cl^{-}(aq)$

At equilibrium : –

$E_{cell} = 0 \Rightarrow E^{0}_{cell} = \frac{RT}{nF}\ell n\, K_{sp}$

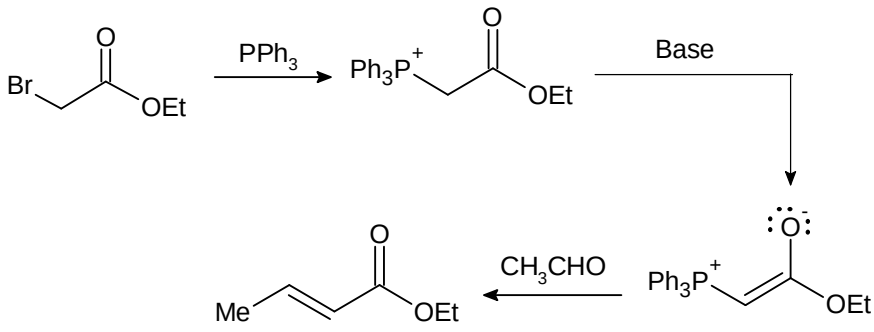
and $E^{0}_{cell} = E^{0}_{Cl^{-}/AgCl/Ag} - E^{0}_{Ag^{+}/Ag}$

$= -0.57V$

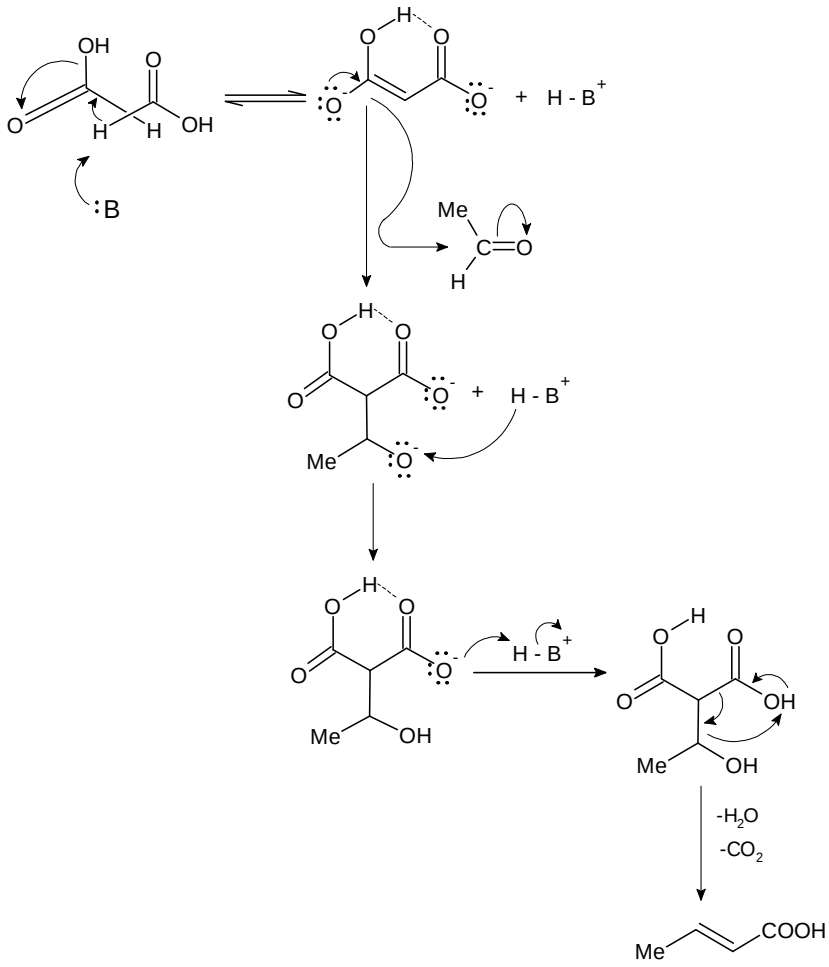
$K_{sp} = 10^{-10} \times \sqrt{10}$

$= 3.16 \times 10^{-10}$

40. (A)



41. (D)



SECTION – C

42. 4

Enantiomeric excess = $\frac{d-\ell}{d+\ell}\times 100 = \frac{68-32}{68+32}\times 100 = 36\%$

$\therefore$ Optical rotation of this mixture = $36\% \times +\left(11.1^0\right)$ = 4

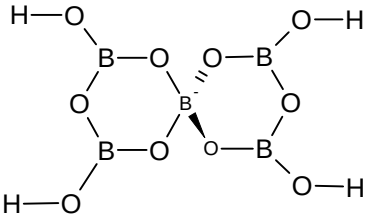
43. 5

Aldol condensation

44. 5

I_3^+, XeF_4, H_2O, NH_2^-, H_2S

45. 6



46. 4

$n = \frac{\lambda}{2\pi r_1} = \frac{1340}{2\times 3.14\times 53} = 4$

Mathematics

PART – III

SECTION – A

47. $f'(x) = \frac{x^6}{6} - \frac{12}{15}x^5 + \frac{5x^3}{20}x^4 + \frac{4x^3}{3} - 2x^2 + 99$

$f''(x) = x^5 - 4x^4 + 3x^3 + 4x^2 - 4x$

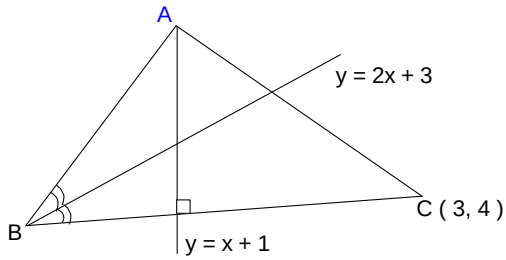
$x\left(x^4 - 4x^3 + 3x^2 + 4x - 4\right)$

$x(x-1)(x-2)^2(x+1)$

Pt of inflexion at X = -1, 0 &1

So -1+0+1 = 0

48. Equation of BC is x + y = 7 so B = $\left(\frac{4}{3}, \frac{17}{3}\right)$



Let slope of AB is m so,

$$\left|\frac{m-2}{1+2m}\right| = \left|\frac{2+1}{1-2}\right|$$

$$m = -\frac{1}{7}$$

So equation of AB is 3x + 21y – 123 = 0

So A ≡ $\left(\frac{17}{4}, \frac{21}{4}\right)$ mid-point of AB = $\left(\frac{67}{24}, \frac{131}{24}\right)$

Equation of medium through C is 7x + y = 25

49. $\left[xy^2dx + \left(y^2dx + 2xydy\right)\right] + x^2ydx + \left(2xydx + x^2dy\right) = 0$

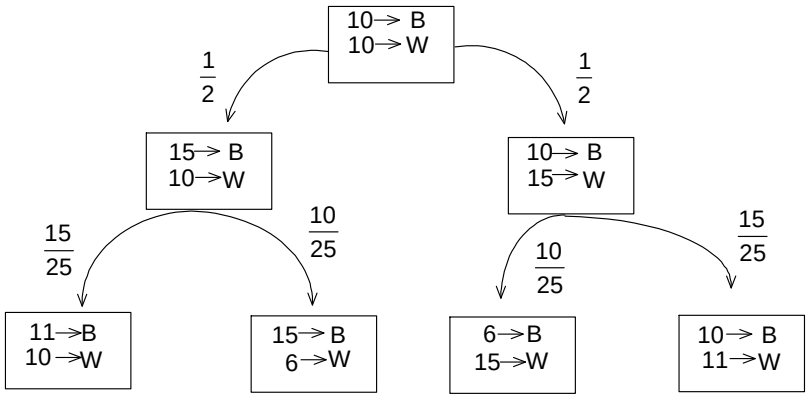
$e^x xy^2dx + e^x\left(y^2dx + xd\left(y^2\right)\right) + e^x x^2ydx + e^x\left(2xydx + x^2dy\right) = 0$

$d\left(e^x xy^2\right) + d\left(e^x x^2y\right) = 0$

$e^x xy^2 + e^x x^2y = C$

Passing through (1, 1) so C = 2e so, curve is $xy(x+y) = 2e^{1-x}$ $xy(x+y) = 2e^{1-x}$

50.

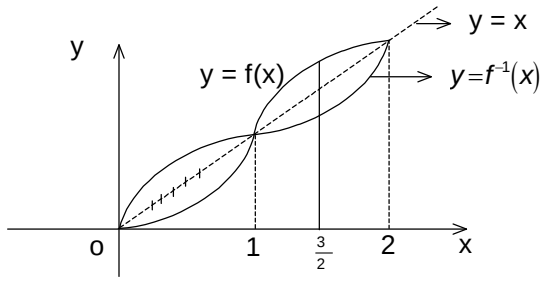


Required probability,

$$\frac{\frac{1}{2}\times\frac{15}{25}\times\frac{10}{21}+\frac{1}{2}\times\frac{10}{25}\times\frac{15}{21}}{\frac{1}{2}\times\frac{15}{25}\times\frac{10}{21}+\frac{1}{2}\times\frac{10}{25}\times\frac{6}{21}+\frac{1}{2}\times\frac{10}{25}\times\frac{15}{21}+\frac{1}{2}\times\frac{15}{25}\times\frac{11}{21}}$$

$$=\frac{2}{1+\frac{2}{5}+1+\frac{11}{10}}=\frac{20}{11+20+4}=\frac{4}{7}$$

51. Required area is $\frac{3}{2}$ times the area of loop



$$\frac{3}{2}\times 2\int_0^1\left[\left(x+x^2\right)-x\right]dx$$

1

52. Let $I=\int_2^4 \times g(x)\,dx$

Using by parts

$$I=x.f^{-1}(x)\Big|_2^4-\int_2^4 f^{-1}(x)\,dx$$

$\rightarrow 1$

$$I=4f^{-1}(4)-2f^{-1}(2)-\int_2^4 f^{-1}(x)\,dx$$

Now,

$f^{-1}(f(x))=x$

$f^{-1}(f(0))=0\Rightarrow f^{-1}(2)=0$

$alsof^{-1}(f(1))=1\Rightarrow f^{-1}(4)=1$

$\int_a^b f(x)\,dx+\int_{f(a)}^{f(b)} f^{-1}(x)\,dx=bf(b)-af(a)$

$\int_0^1 f(x)\,dx+\int_2^4 f^{-1}(x)\,dx=1.f(1)-0.f(0)$

$\int_0^1\left(x^3+x+\sin\pi x+2\right)dx+\int_2^4 f^{-1}(x)\,dx=4$

$$also\ \int_2^4 f^{-1}(x)\,dx=4-\left(\frac{x^4}{4}+\frac{x^2}{2}-\frac{\cos\pi x}{\pi}+2x\right)_0^1$$

53. Let x = cosθ

$$x\rightarrow\frac{1}{\sqrt{2}}+,\theta\rightarrow\frac{\pi}{4}^-$$

$$\text{As}\ \lim_{\theta\rightarrow\frac{\pi}{4}^-}\frac{\cos^{-1}\cos 2\theta}{\cos\theta-\frac{1}{\sqrt{2}}}=\lim_{\theta\rightarrow\frac{\pi}{4}^-}\frac{\frac{\pi}{2}-2\theta}{\cos\theta-\frac{1}{\sqrt{2}}}$$

$$\lim_{\theta\rightarrow\frac{\pi}{4}^-}\frac{-2}{-\sin\theta}=2\sqrt{2}$$

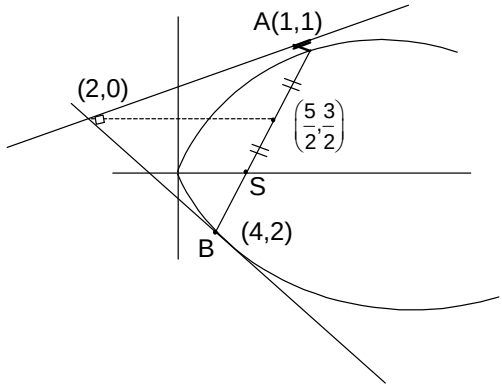
54. Coeff. X^{3r} in $\left[\left(1+x+x^2\right)^n.(1-x)^n\right]$

$\Rightarrow coeff.of\ x^{3r}\ in\left[\left(a_0+a_1x+.....a_2x^{2n}\right)\left(n_{c_0}-n_{c_1}\times.....\right)\right]$

$={}^nC_r$

55. $\because$ Tangents are $\perp r$. so, they intersect on directrix.

Point of intersection = (2,0) mid-point of (1,1) & (4,2) is $\left(\frac{5}{2},\frac{3}{2}\right)$



Slope of axis = $\frac{\frac{3}{2}-0}{\frac{5}{2}-2}=3$

Equation of directrix, $y=-\frac{1}{3}(x-2)$

$x+3y=2$

AB is focal chord,

BS = ($\perp_r$ distance from B on directrix) = $\frac{4+6-2}{\sqrt{10}}=\frac{8}{\sqrt{10}}$

AS = ($\perp_r$ distance from A on directrix) = $\frac{1+3-2}{\sqrt{10}}=\frac{2}{\sqrt{10}}$

So, focus divides AB in 1: 4 ratios. So S = $\left(\frac{8}{5},\frac{6}{5}\right)$

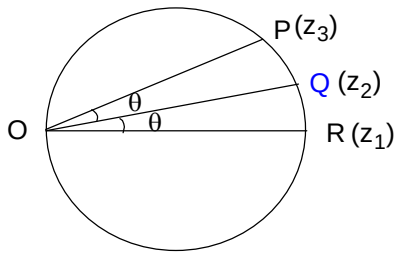
56. By Rotation theory

$\frac{z_2}{z_1}=\cos\theta e^{i\theta}$

$\frac{z_3}{z_2}=\frac{\cos2\theta}{\cos\theta}.e^{i\theta}$

$so\ \frac{z_2^{\ 2}}{z_3z_1}=\frac{\cos^2\theta}{\cos2\theta}=\frac{3}{2}$

$\theta=\frac{\pi}{6}$



57. $\Delta AXE\simeq\Delta CYE$

$so, ar\left(\Delta AXE\right)=ar\left(\Delta CYE\right)=\Delta_1$

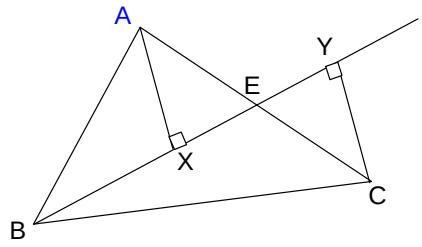
$ar\left(\Delta BYC\right)=ar\left(\Delta BEC\right)+\Delta_1$

$4\Delta_1=ar\left(\Delta BEC\right)+\Delta_1$

$ar\left(\Delta BEC\right)=3\Delta_1=ar\left(\Delta ABE\right)=ar\left(\Delta AXB\right)+ar\left(\Delta AXE\right)$

$\Rightarrow so, ar\left(\Delta AXB\right)=2\Delta_1$

$ar\left(\Delta ABC\right)=2ar\left(\Delta BEC\right)=6\Delta_1$



58. $\frac{A}{a},\frac{B}{b},\frac{C}{c}HP$

$\Rightarrow \frac{2b}{B} = \frac{a}{A} + \frac{c}{C}$

$\Rightarrow 2bB = aC + cA$

$\Rightarrow aB + cB = aC + cA$

$\Rightarrow a[B - C] = c[A - B]$

$so, r = \frac{c}{a}$

$\frac{A^2}{a}, \frac{B^2}{b}, \frac{C^2}{c} are in HP \Rightarrow r^2 = \frac{c}{a}$

59 – 60. A ≡ (3, 2, 1), B ≡ (4, 1, 2) point of intersection ≡ (0, 0, 0)

Area of triangle = $\frac{1}{2}\left|(3\hat{i} + 2\hat{j} + \hat{k}) \times (4\hat{i} + \hat{j} + 2\hat{k})\right| = \sqrt{\frac{19}{2}}$

Also image of (0, 0, 0) w.r.t plane π is $\left(\frac{14}{3}, \frac{7}{3}, \frac{-7}{3}\right)$

Image of (0, 0, 0) w.r.t AB is $\left(\frac{14}{3}, \frac{16}{3}, \frac{2}{3}\right)$

Equation of line joining these two points = $\frac{3x - 14}{0} = \frac{3y - 7}{3} = \frac{3z + 7}{3}$

61 – 62. $\sum_{i=1}^{n-1}\left(x_i - 2i\sqrt{x_i - i^2}\right) = 0$

$\sum_{i=1}^{n-1}\left(\sqrt{x_i - i^2}\right)^2 - 2i\sqrt{x_i - i^2} + i^2 = 0$

$\sum_{i=1}^{n-1}\left(\sqrt{x_i - i^2} - i\right)^2 = 0$

$so, x_i = 2i^2$

Now, $x_1^2 + \ldots\ldots + x_n^2 = 280$

$2\left[1^2 + 2^2 + \ldots\ldots n^2\right] = 280$

$n = 7$

$y_1 + y_2 + y_3 = 7$

$y_1^1 + y_2^1 + y_3^1 = 4$

${}^{4+3-1}C_3 = {}^6C_3 = 20$

Total triangles formed = ${}^{15}C_3 = \frac{15 \times 14 \times 13}{6}$

N of isosceles triangles formed = 15 × 7

probalality = $\frac{15 \times 7}{15 \times 14 \times 13} \times 6$

$\frac{3}{13}$

63 – 64. p (n) = p(n - 1) + p(n - 2)

SECTION – C

65. Consider $y = ax^2 + bx + c$

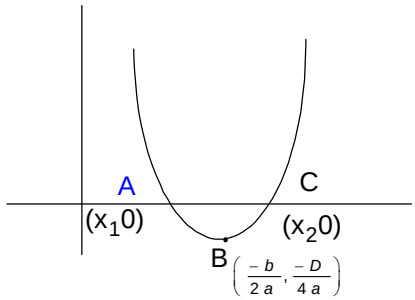
Slope of BA. Slope of BC = -1

$$\frac{\frac{D}{4a}}{x_1 + \frac{b}{2a}} \cdot \frac{\frac{D}{4a}}{x_2 + \frac{b}{2a}} = -1$$

$$\Rightarrow \frac{D^2}{16a^2} + \frac{b^2}{4a^2} + (x_1 + x_2)\frac{b}{2a} + x_1x_2 = 0$$

$$\Rightarrow \frac{D^2}{16a^2} + \frac{b^2}{4a^2} - \frac{b}{a}\left(\frac{b}{2a}\right) + \frac{c}{a} = 0$$

$$\Rightarrow D = 0 \text{ or } 4$$



66. $$I = \int_{-\pi}^{\pi} \sqrt{\frac{|\sin x|}{1+\tan^2 x}}\,dx$$

$$I = 2\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \sqrt{\frac{\sin x}{1+\cot^2 x}}\,dx$$

$$= 4\int_{0}^{\frac{\pi}{2}} \sqrt{\cos x}\sin x\,dx$$

$$= \frac{8}{3}$$

$$[I] = 2$$

67. PA = PR = 8 – a

So PQ = 8 – a + b

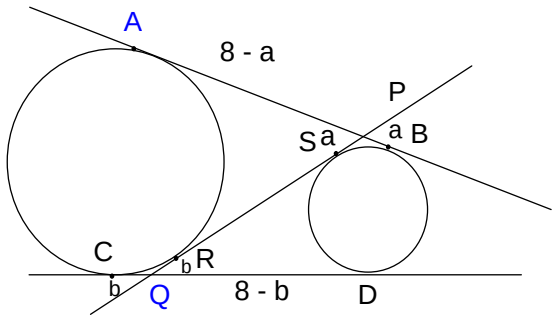
Also QD = QR = 8 – b

So QP = 8 – b + a

8 – a + b = 8 – b + a

$\Rightarrow a = b$

$PQ = 8$



68. $\det\left(A^{\mu-1}(A-5I)\right) = \det(A^{\mu-1}).\det(A-51)$

69. Tangent at $\left(\sqrt{5}, \frac{4}{3}\right)$ is $\sqrt{5}x + 3y = 9$ x=1 is directrix of Hyperbola.

So, A = $\left(1, \frac{9-\sqrt{5}}{3}\right)$ & B = (0, 3)

Area of figure OGAB is

$$1\times\left(\frac{9-\sqrt{5}}{3}\right)+\frac{1}{2}\times 1\times\left(3-\frac{9-\sqrt{5}}{3}\right)$$

$$=\frac{9-\sqrt{5}}{3}+\frac{\sqrt{5}}{6}$$

$$=\frac{18-\sqrt{5}}{6}$$

Area of required figure is

$$4\left\{\frac{18-\sqrt{5}}{6}\right\}=\frac{36-2\sqrt{5}}{3}$$

Integral part of area is 7.

